Lab 11: Synthetic Control

SOCIOL 723 — Social Statistics II — Thursday, November 5

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1 Goals

1. Construct a synthetic control and read the weight table.
2. Judge the design by pre-treatment fit, not by the size of the gap.
3. Run placebo inference in space and in time.
4. Check robustness by leaving donors out.

```r
library(dplyr)
library(ggplot2)
library(tidysynth)

data("smoking", package = "tidysynth")
```

```r
c(rows = nrow(smoking), states = n_distinct(smoking$state),
  years = paste(range(smoking$year), collapse = "-"))
#>        rows      states       years
#>      "1209"        "39" "1970-2000"
```

2 The design

Question. What did California's Proposition 99 (1988) do to cigarette consumption?

Treated unit. California. **Donor pool.** 38 states that adopted no comparable programme. **Outcome.** Per-capita cigarette sales. **Pre-period.** 1970–1988.

> **Note**
>
> The donor pool was restricted *on substantive grounds before estimating*: states with their own large tobacco taxes or control programmes are excluded, because they are treated too. That decision is part of the design, not a robustness check.

```r
smoking |>
  mutate(ca = state == "California") |>
  ggplot(aes(year, cigsale, group = state, colour = ca, linewidth = ca)) +
  geom_line(alpha = .8) +
  scale_colour_manual(values = c("grey78", "firebrick"), guide = "none") +
  scale_linewidth_manual(values = c(.3, 1.1), guide = "none") +
  geom_vline(xintercept = 1988, linetype = 2) +
  labs(x = NULL, y = "packs per capita") +
  theme_minimal(base_size = 10)
```

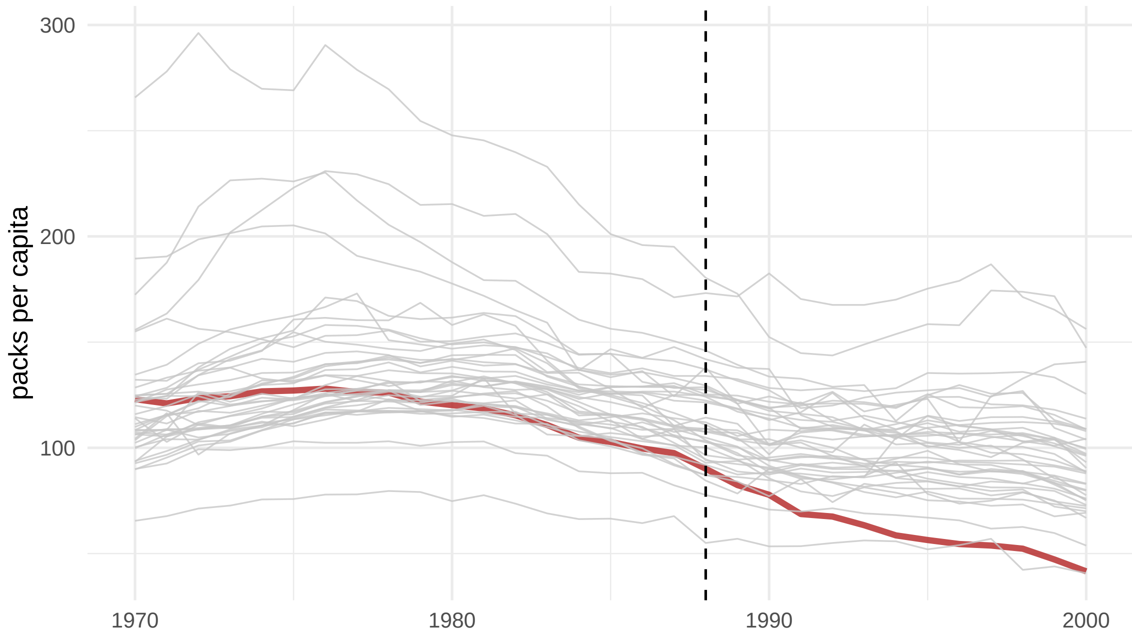

Figure 1: Raw trajectories. California is one line among 39; the eye cannot pick out a treatment effect.

3 Building the synthetic control

```
sc <- smoking |>
  synthetic_control(outcome = cigsale, unit = state, time = year,
                    i_unit = "California", i_time = 1988,
                    generate_placebos = TRUE) |>
  # predictors measured over the pre-period
  generate_predictor(time_window = 1980:1988,
                     ln_income = mean(lnincome, na.rm = TRUE),
                     ret_price = mean(retprice, na.rm = TRUE),
                     youth     = mean(age15to24, na.rm = TRUE)) |>
  generate_predictor(time_window = 1984:1988,
                     beer_sales = mean(beer, na.rm = TRUE)) |>
  # lagged outcomes: these do most of the work
  generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
  generate_predictor(time_window = 1980, cigsale_1980 = cigsale) |>
  generate_predictor(time_window = 1988, cigsale_1988 = cigsale) |>
  generate_weights(optimization_window = 1970:1988) |>
  generate_control()
```

3.1 Who is synthetic California?

```
sc |> grab_unit_weights() |> filter(weight > 0.001) |>
  arrange(desc(weight)) |> knitr::kable(digits = 3)
```

unit	weight
Utah	0.343
Nevada	0.236
Montana	0.182
Colorado	0.175
Connecticut	0.062

```
sc |> grab_predictor_weights() |> arrange(desc(weight)) |> knitr::kable(digits = 3)
```

variable	weight
cigsale_1975	0.493
cigsale_1980	0.392
cigsale_1988	0.068
ret_price	0.031
beer_sales	0.012
youth	0.003
ln_income	0.001

> **!** Important
>
> **Print the weight table.** It is the single most informative output of a synthetic control study: readers can decide for themselves whether a weighted average of these states is a plausible California. No regression coefficient offers that.

3.2 Balance

```
sc |> grab_balance_table() |> knitr::kable(digits = 2)
```

variable	California	synthetic_California	donor_sample
ln_income	10.08	9.86	9.83
ret_price	89.42	89.32	87.27
youth	0.17	0.17	0.17
beer_sales	24.28	24.10	23.66
cigsale_1975	127.10	126.90	136.93
cigsale_1980	120.20	120.25	138.09

variable	California	synthetic_California	donor_sample
cigsale_1988	90.10	91.43	113.82

4 The result

```
plot_trends(sc)
```

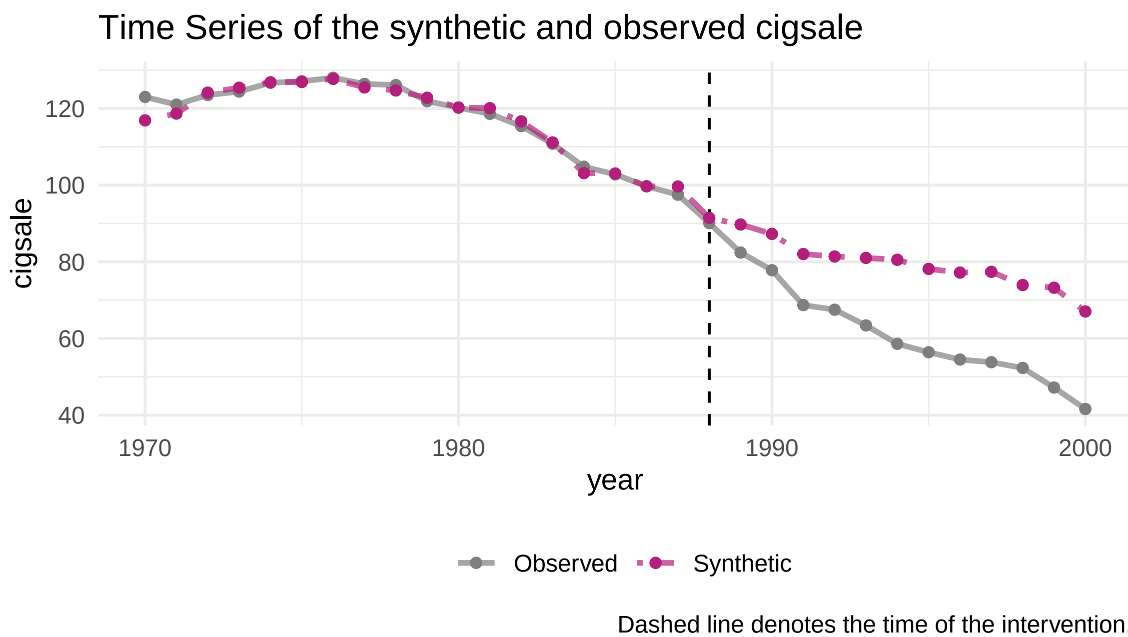

Figure 2: California and synthetic California. The pre-1988 lines nearly coincide, which is what licenses reading the post-1988 gap as an effect.

```
plot_differences(sc)
```

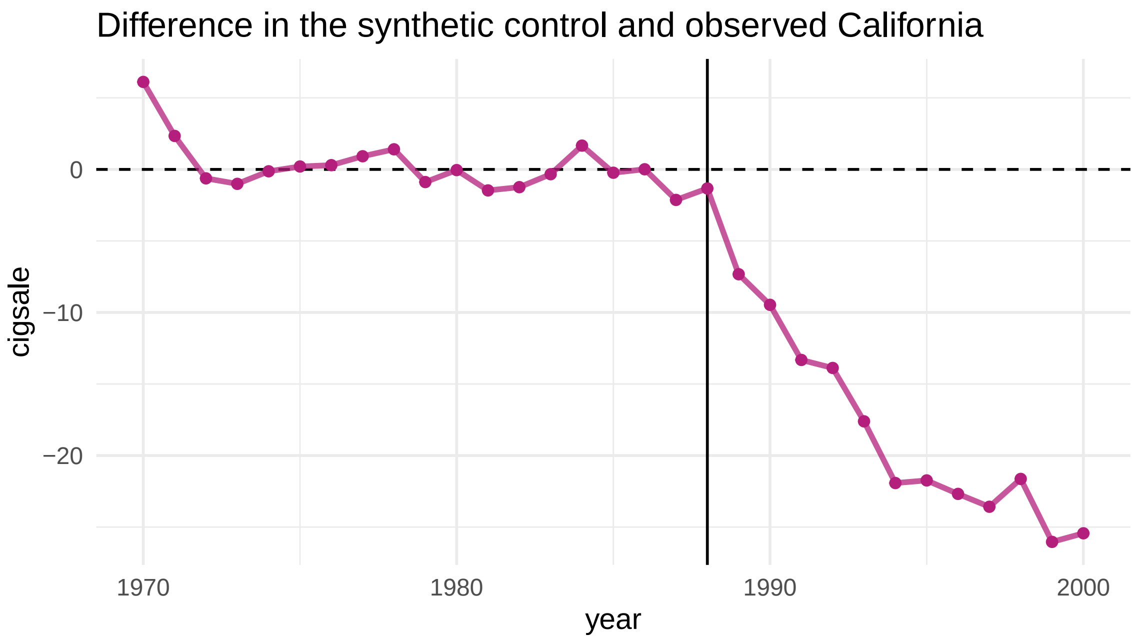

Figure 3: The gap between California and its synthetic control.

```
gaps <- sc |> grab_synthetic_control() |>
  mutate(gap = real_y - synth_y)
c(mean_gap_post = mean(gaps$gap[gaps$time_unit > 1988]),
  gap_2000 = gaps$gap[gaps$time_unit == 2000],
  rmspe_pre = sqrt(mean(gaps$gap[gaps$time_unit <= 1988]^2)))
#> mean_gap_post      gap_2000     rmspe_pre
#>       -18.719       -25.435         1.791
```

The pre-treatment RMSPE is the number to look at first. If it were large relative to the post-treatment gap, there would be nothing to report.

5 Inference

5.1 Placebos in space

```
plot_placebos(sc)
```

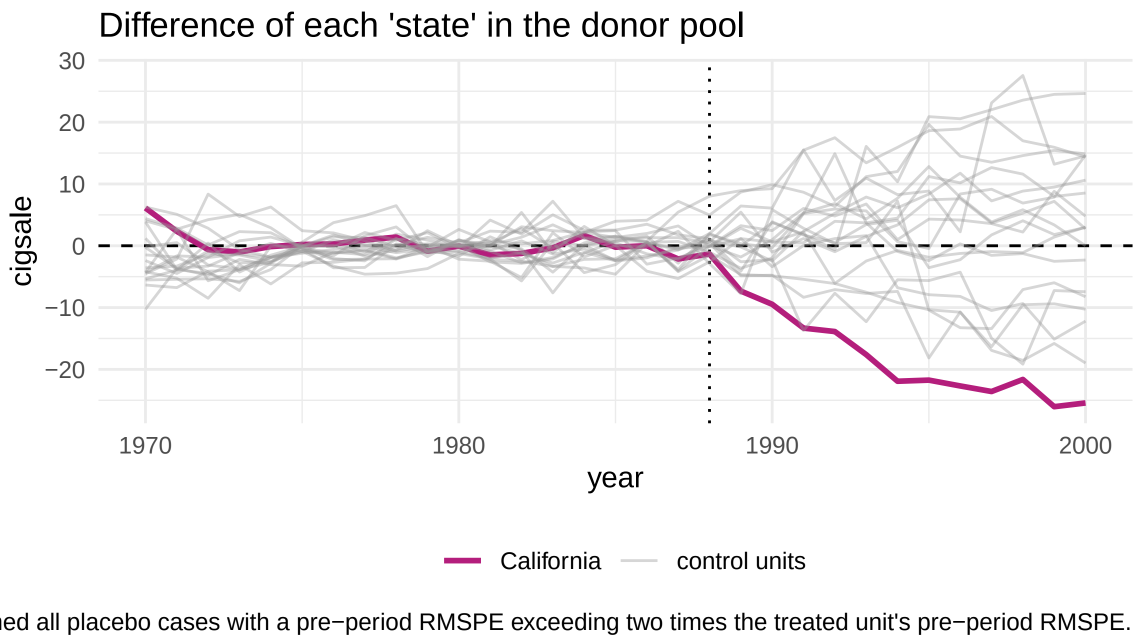

Figure 4: Each grey line is a donor state treated as if it had adopted Proposition 99. California is in black.

`plot_placebos()` drops donors whose pre-treatment fit is much worse than California's — otherwise states the method could never match would dominate the picture for reasons unrelated to treatment.

5.2 The RMSPE ratio

```
plot_mspe_ratio(sc)
```

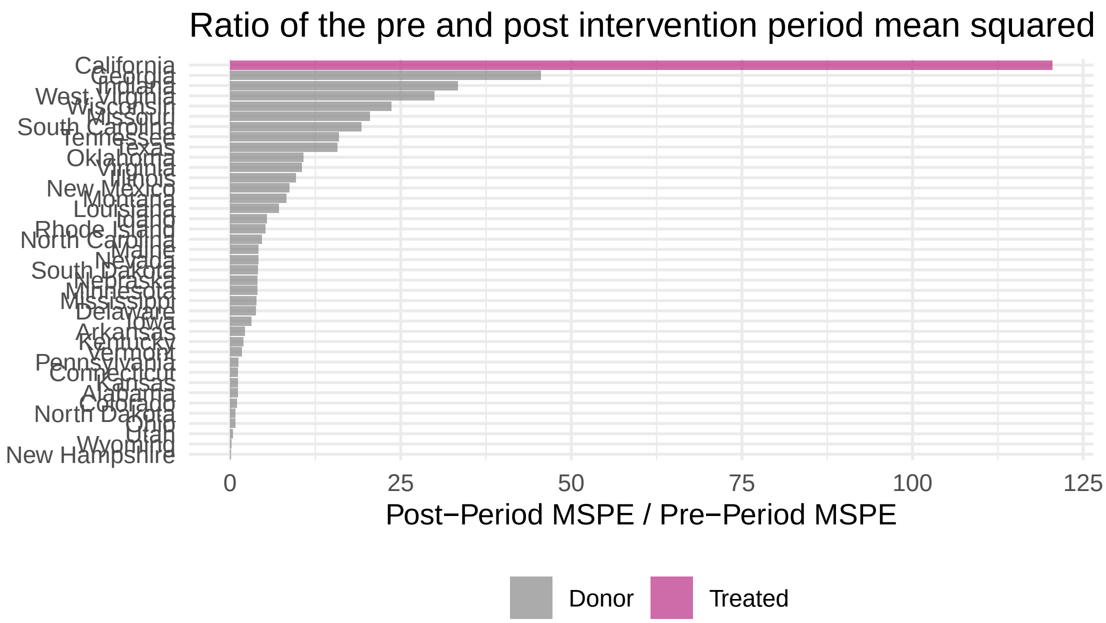

Figure 5: Ratio of post- to pre-treatment RMSPE. California is the largest of all 39 units.

```
sig <- sc |> grab_significance()
sig |> select(unit_name, type, mspe_ratio, rank, fishers_exact_pvalue) |>
  head(6) |> knitr::kable(digits = 3)
```

unit_name	type	mspe_ratio	rank	fishers_exact_pvalue
California	Treated	120.49	1	0.026
Georgia	Donor	45.53	2	0.051
Indiana	Donor	33.37	3	0.077
West Virginia	Donor	29.93	4	0.103
Wisconsin	Donor	23.61	5	0.128
Missouri	Donor	20.47	6	0.154

The `fishers_exact_pvalue` column is the unit's rank divided by the number of units. With 38 donors the smallest attainable value is $1/39 \approx 0.026$ — a hard floor worth stating when you report it. Strictly this is a *placebo rank* rather than an exact test: exactness would require treatment to have been randomly assigned among these 39 states.

5.3 A placebo in time

Pretend the programme began in 1980, eight years early. Nothing should happen.

```
sc_1980 <- smoking |>
  filter(year <= 1988) |>
  synthetic_control(outcome = cigsale, unit = state, time = year,
                    i_unit = "California", i_time = 1980,
```

```
                    generate_placebos = FALSE) |>
  generate_predictor(time_window = 1970:1980,
                     ln_income = mean(lnincome, na.rm = TRUE),
                     ret_price = mean(retprice, na.rm = TRUE)) |>
  generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
  generate_weights(optimization_window = 1970:1980) |>
  generate_control()

g80 <- sc_1980 |> grab_synthetic_control() |> mutate(gap = real_y - synth_y)
c(mean_gap_1981_1988 = mean(g80$gap[g80$time_unit > 1980]),
  rmspe_pre = sqrt(mean(g80$gap[g80$time_unit <= 1980]^2)))
#> mean_gap_1981_1988          rmspe_pre
#>             -12.85               4.57
```

A gap near zero after the fake intervention is reassuring. A large one would mean the method manufactures effects.

5.4 Leave-one-donor-out

```
donors <- sc |> grab_unit_weights() |> filter(weight > 0.01) |> pull(unit)

## refit the EXACT original pipeline after dropping each donor, so that any
## change is attributable to the donor and not to a specification change
loo <- lapply(donors, function(dd) {
  s <- smoking |> filter(state != dd) |>
    synthetic_control(outcome = cigsale, unit = state, time = year,
                      i_unit = "California", i_time = 1988,
                      generate_placebos = FALSE) |>
    generate_predictor(time_window = 1980:1988,
                       ln_income = mean(lnincome, na.rm = TRUE),
                       ret_price = mean(retprice, na.rm = TRUE),
                       youth     = mean(age15to24, na.rm = TRUE)) |>
    generate_predictor(time_window = 1984:1988,
                       beer_sales = mean(beer, na.rm = TRUE)) |>
    generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
    generate_predictor(time_window = 1980, cigsale_1980 = cigsale) |>
    generate_predictor(time_window = 1988, cigsale_1988 = cigsale) |>
    generate_weights(optimization_window = 1970:1988) |>
    generate_control()
  g <- s |> grab_synthetic_control() |> mutate(gap = real_y - synth_y)
  tibble::tibble(dropped = dd, mean_gap = mean(g$gap[g$time_unit > 1988]))
}) |> bind_rows()
```

```r
knitr::kable(loo, digits = 2)
```

dropped	mean_gap
Colorado	-19.26
Connecticut	-19.19
Montana	-17.89
Nevada	-19.31
Utah	-17.64

If dropping any single donor moved the estimate substantially, the result would rest on one state rather than on a comparison group.

6 Exercises

1. **Predictor sensitivity.** Re-estimate using only lagged outcomes as predictors (drop income, price, beer, youth). How much do the weights and the estimate move? What does that tell you about how much work the covariates are doing?
2. **Pre-period length.** Re-estimate using only 1980–1988 to fit the weights. Does the pre-treatment fit worsen? Does the estimate change? Connect this to the factor-model justification from lecture.
3. **A bad donor pool.** Deliberately include a state with its own tobacco programme. How does that contaminate the comparison, and would any diagnostic catch it?
4. **DID for comparison.** Estimate a plain 2×2 DID comparing California to the unweighted average of all donors, before and after 1988. How does it differ from the synthetic control estimate, and why?
5. **A different treated unit.** Using `set.seed(723)`, pick a donor state at random, treat it as treated, and run the whole pipeline. Write the paragraph you would write if this were your result — then explain what stops you from believing it.

7 Session information

```r
sessionInfo()
#> R version 4.4.1 (2024-06-14)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS 26.5.2
#>
#> Matrix products: default
#> BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;
```

```
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: Asia/Shanghai
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] tidysynth_0.2.1 ggplot2_4.0.0   dplyr_1.1.4
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6       jsonlite_1.8.9      compiler_4.4.1
#>  [4] tidyselect_1.2.1   tinytex_0.53        tidyr_1.3.1
#>  [7] scales_1.4.0       yaml_2.3.10         fastmap_1.2.0
#> [10] R6_2.5.1           labeling_0.4.3      generics_0.1.3
#> [13] kernlab_0.9-33     knitr_1.48          forcats_1.0.0
#> [16] tibble_3.2.1       nloptr_2.1.1        pillar_1.9.0
#> [19] RColorBrewer_1.1-3 rlang_1.3.0         utf8_1.2.4
#> [22] xfun_0.47          S7_0.2.0            cli_3.6.5
#> [25] withr_3.0.1        magrittr_2.0.3      digest_0.6.37
#> [28] grid_4.4.1         lifecycle_1.0.4     vctrs_0.6.5
#> [31] pracma_2.4.4       evaluate_1.0.0      glue_1.7.0
#> [34] farver_2.1.2       numDeriv_2016.8-1.1 optimx_2024-12.2
#> [37] fansi_1.0.6        rmarkdown_2.28      purrr_1.0.2
#> [40] tools_4.4.1        pkgconfig_2.0.3     htmltools_0.5.8.1
```